

# Enzyme Action: Preparing the pH Investigation

LAUNCH • Science • 40 minutes

I can explain enzyme specificity and denaturation, then identify the variables and evidence needed to investigate the effect of pH on amylase safely.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

You may write, type, say, sign, point or direct a partner. An adult can read the words and record your exact response. Keep the choice and explanation your own.

## Words for this lesson

| Word        | Meaning                     |
|-------------|-----------------------------|
| specificity | one substrate fits          |
| denature    | active site changes shape   |
| variable    | what changes or is measured |

## First idea

Three amylase mixtures contain the same starch and are kept at the same temperature. Which first prediction is scientifically testable?

- A. Amylase will always work equally fast because enzymes never change.
- B. There may be an optimum pH where starch disappears in the shortest time.
- C. The buffer with the brightest label will make the reaction fastest.

My first idea and reason:



Three tubes all blue-black with iodine at the start

## We do • choose, explain, check

Which statement best explains enzyme specificity?

- A. Every enzyme binds every substrate if enough time passes.
- B. An enzyme's active site has a complementary shape to its particular substrate.
- C. The substrate permanently changes the enzyme into a product.

My choice: \_\_\_\_\_ Evidence or reason:

## Variables and validity planning console

Route each planning card to independent variable, dependent evidence or control/validity before writing the method.

Read or hear one card. Choose a group. Give the evidence. A partner checks your reason before you move on. Point or write card numbers; cutting is optional.

| Group                | Card numbers and evidence |
|----------------------|---------------------------|
| INDEPENDENT VARIABLE |                           |
| DEPENDENT EVIDENCE   |                           |
| CONTROL / VALIDITY   |                           |

## Activity cards

|                                                                                                                    |                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>1 • Buffer pH</b><br>Use labelled buffers so mixtures differ in pH according to the centre-approved method.     | <b>2 • Endpoint time</b><br>Record the time until a sample no longer makes iodine turn blue-black.               |
| <b>3 • Constant temperature</b><br>Keep every reaction at the same approved temperature throughout the comparison. | <b>4 • Equal volumes</b><br>Use the same approved volume and concentration of amylase and starch in every trial. |

**5 • Sampling interval**

Sample every fixed number of seconds using the interval in the centre method.

**6 • No-amylase control**

A separate starch mixture contains no active amylase and should remain starch-positive.

State the predicted causal chain: changing pH may change active-site shape, which changes endpoint time and calculated rate.

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## Your lesson record

Produce a safe, ordered CP 1.10 planning record that another trained person could follow using the centre-approved Pearson method and local risk assessment.

### Model helps us see

Use the model to connect active-site shape, substrate binding and a measurable endpoint. Record the sentence: pH may affect active-site shape, so we measure the time until starch is no longer detected.

### Model does not show

Card shapes do not show molecular movement, collisions, bonds or gradual changes in protein structure. A neat lock-and-key picture is a simplified representation, not a literal rigid lock.

### Supported

Complete a labelled IV/DV/CV table and order six picture-method cards; finish one because sentence about keeping temperature constant.

Use the word bank and sentence stem: We change \_\_\_. We measure \_\_\_. We keep \_\_\_ the same because \_\_\_.

### Standard

Write an ordered method outline, identify variables and control sample, and design a table for pH, endpoint time and rate.

Use numbered steps and the command-word frame: EXPLAIN = point + scientific reason.

### Stretch

Justify the controls and sampling interval, predict the rate pattern, and identify one source of uncertainty before practical work begins.

Distinguish validity, precision and safety; do not invent an unsupported concentration or heating method.

### Evidence target

A pupil-owned prediction, IV/DV/CV table, ordered method outline, results-table headings and one justified safety/control decision.

### Exit, Audience and evidence • W8L1

Supported: Complete: amylase acts on \_\_\_. We change \_\_\_ and measure \_\_\_.

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Standard: Explain why temperature and volumes must be controlled when investigating pH and amylase.

Stretch: Justify the no-amylase control and distinguish it from a control variable.

**What I said, and what it changed**

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**Next I will \_\_\_\_\_.**

**Adult witness note**

What was observed, and with what level of independence?

**My evidence and explanation**

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What does this evidence not show?

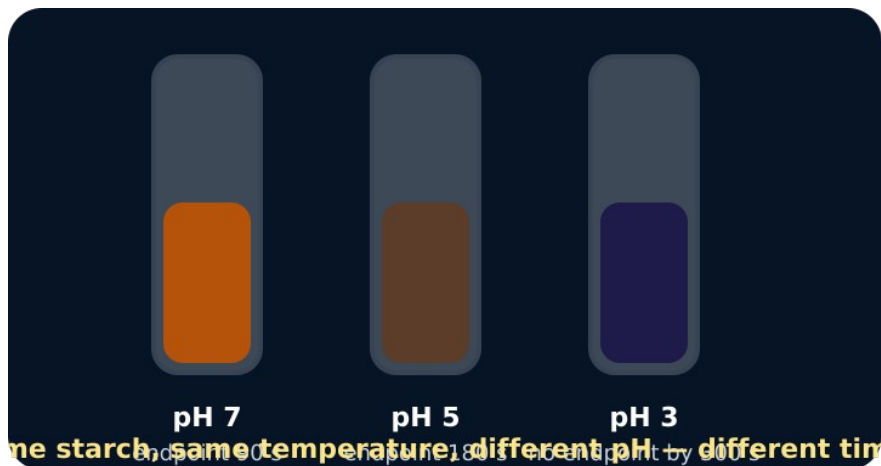
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## Visual reference cards

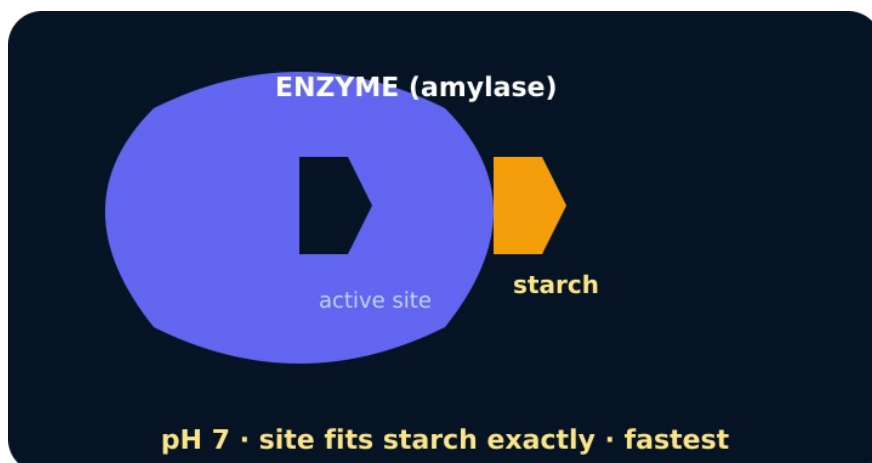
These are diagrams or model views from the uploaded lesson. Use their labels and the card descriptions; they are not photographs of your own results.



One tube orange-brown, two still blue-black



Three tubes at different pH reaching the endpoint at different times



Enzyme active site model at the chosen pH

### Exit • one next step

After the adult has listened, what will you keep, improve or check next?

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